

engage in valid Bitcoin transactions. This iteration finds its primary utility in the domains of currency and payments.

The overarching goal of Blockchain Version 1.0, also referred to as Blockchain Version 1.0, was to usher in a transparent, publicly accessible, fully decentralised, immutable ledger, and a distributed transaction system within the global financial landscape. It takes its foundational cues from the concept and structure of Bitcoin, with a primary emphasis on the genesis of novel cryptocurrencies. Blockchain 1.0 is often described as a digital, decentralised, distributed ledger that meticulously records transactions within a universally shared database. These records are updated by blockchain miners, and collectively overseen and maintained by all participants, without any single entity claiming ownership.

Blockchain 2.0 (smart contracts)

The introduction of Blockchain Version 2.0 stemmed from inherent issues found in its predecessor, Version 1.0. Notably, Version 1.0 faced challenges related to the resource-intensive nature of Bitcoin mining and a notable lack of network scalability. Version 2.0 was conceived as a means to rectify these shortcomings.

In this upgraded iteration, blockchain extends its functionality beyond merely handling cryptocurrencies to encompass smart contracts. These smart contracts can be likened to compact computer programs residing within the interconnected blocks of the chain. They autonomously execute predefined conditions, such as facilitation, verification, or enforcement, thus streamlining processes and enhancing transaction cost efficiency.

One significant shift in Blockchain 2.0 involved replacing Bitcoin with Ethereum as its foundational cryptocurrency. This transition proved highly successful, enabling Blockchain 2.0 to efficiently process a substantial volume of transactions on the public network with remarkable speed.

Blockchain 3.0 (DApps)

Version 3.0, that followed Version 2.0, features the inclusion of DApps, commonly known as decentralised applications. These DApps resemble traditional applications in that they can have a frontend coded in any language, facilitating communication with their backend. However, what sets them apart is that their backend operations are executed within a decentralised peer-to-peer network. They harness decentralised storage and communication mechanisms, such as Ethereum Swarm, to function.

DApps are inherently decentralised, devoid of a single owner or central authority, ensuring several key advantages, including transparency, heightened security, universal data accessibility, absence of censorship, and a flexible development environment. The adoption of DApps brings forth numerous benefits, including uninterrupted operation (zero downtime), robust privacy protection, data integrity assurance, and trustworthy, yet secure, communication for various purposes such as business and transactions.

Why is blockchain important?

Transparency and trust: Blockchain ensures transparency and an immutable record of transactions, making it difficult to alter or manipulate data. This transparency and immutability build trust among participants and eliminate the need for intermediaries, reducing costs and increasing efficiency.

Security: Blockchain uses cryptographic algorithms to secure transactions and data. The decentralised nature of blockchain networks makes them resilient to hacking and fraud attempts. This enhanced security is crucial for applications like financial transactions, supply chain management, and identity verification.

Decentralisation: Blockchain operates on a distributed network

of computers or nodes, eliminating the need for a central authority or intermediary. This decentralisation ensures that no single entity has control over the network and reduces the risk of single points of failure or censorship.

Innovation and disruption:

Blockchain technology has the potential to disrupt traditional industries and business models by introducing new ways of transacting, storing data, and executing contracts. It enables the development of decentralised applications, smart contracts, and new financial instruments, fostering innovation and opening up new business opportunities.

How does blockchain work?

The developments in blockchain are relatively nascent, yet they harbour the potential to bring about revolutionary changes in the future. So, let's start the process of analysing this technology as shown in Figure 2.

Blockchain comprises a synergy of three key technologies:

1. Cryptographic keys
2. A peer-to-peer network housing a shared ledger
3. A computational framework for storing the network's transactions and records

Cryptography keys, comprising the private and public keys, are foundational to blockchain. These keys play a crucial role in facilitating secure and prosperous transactions between two parties. Every individual possesses both these keys, deploying them to establish a robust and unassailable digital identity reference, an element of paramount significance within the blockchain framework.

In the realm of cryptocurrency, this digital identity takes the guise of a 'digital signature', a term widely employed to authorise and oversee transactions. The digital signature seamlessly integrates into the peer-to-peer network, where a diverse